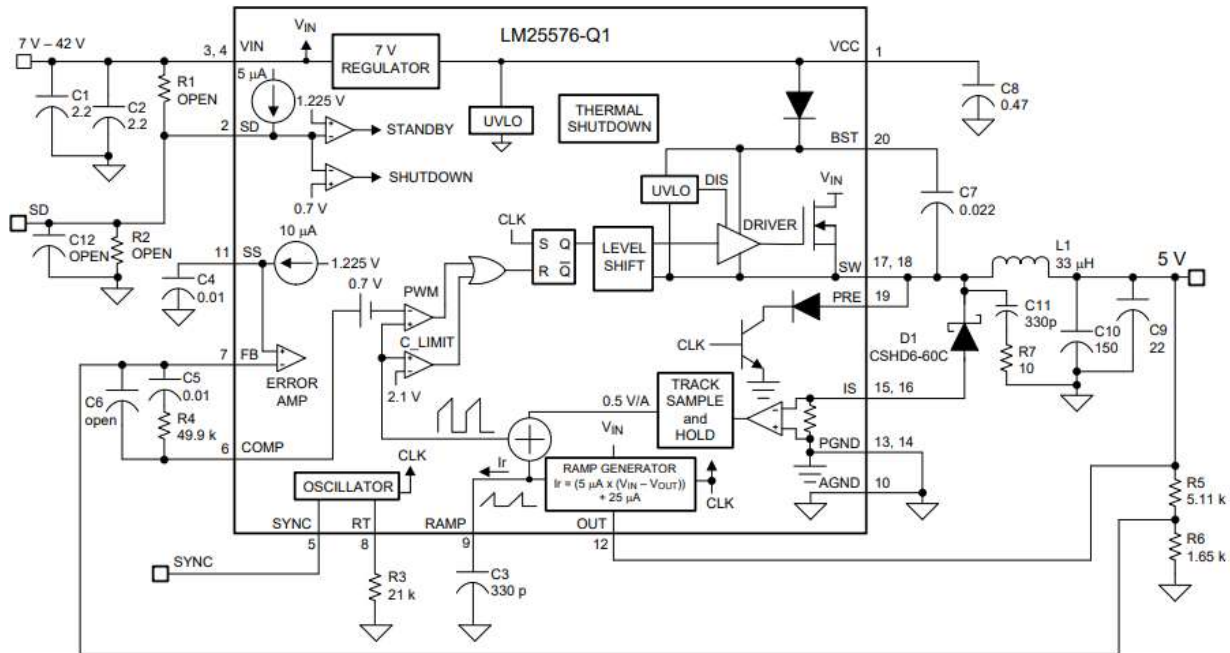


```

% Step-down switch converter
% Vin = 6V - 8.4V
% Vout = 3.3V 2-3A
% Fs= 500kHz
% Minimum load current (for CCM) = 250mA (was 100mA)
% Maximum Load Current = 3A

```



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```

% Calculation of RT

```

```

% 580e-9 and 135e-12 are constants from internal oscillator timing
% characteristics

```

$$RT = ((1/(500e3)) - 580e-9) / (135e-12)$$

$$RT = 1.0519e+04$$

```

% answer is 10.5k, nearest e24 = 10k (5%), e96 = 10.5k (1%)

```

$$F_{swe24} = 1 / (580e-9 + (135e-12) * 10.51e3) \quad \% \text{ im buying e24 smd resistors,}$$

$$F_{swe24} = 5.0029e+05$$

```

% adafruit 441 book has 10k + 510 = 10.51 ----- do this -----

```

$$F_{swe96} = 1 / (580e-9 + (135e-12) * 10.5e3)$$

$$F_{swe96} = 5.0063e+05$$

```

% picture example below

```

$$R_T = \frac{[(1 / 300 \times 10^3) - 580 \times 10^{-9}]}{135 \times 10^{-12}}$$

```

% this section is for calculating the inductor value

```

```
% inductor ripple must be smaller than 2*iout,min
twoXiout= 2*0.100
```

```
twoXiout = 0.2000
```

8.1.3 L1

The inductor value is determined based on the operating frequency, load current, ripple current, and the minimum and maximum input voltage ($V_{IN(min)}$, $V_{IN(max)}$):

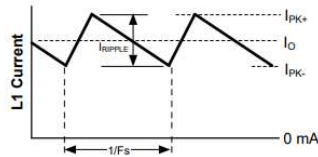


Figure 14. Inductor Current Waveform

To keep the circuit in continuous conduction mode (CCM), the maximum ripple current I_{RIPPLE} should be less than twice the minimum load current, or 0.5 A p-p. Using this value of ripple current, the value of inductor (L1) is calculated using the following:

$$L1 = \frac{V_{OUT} \times (V_{IN(max)} - V_{OUT})}{I_{RIPPLE} \times F_S \times V_{IN(max)}} \quad (6)$$

$$L1 = \frac{5V \times (42V - 5V)}{0.5A \times 300 \text{ kHz} \times 42V} = 29 \mu H \quad (7)$$

This procedure provides a guide to select the value of L1. The nearest standard value (33 μH) will be used. L1 must be rated for the peak current (I_{PK+}) to prevent saturation. During normal loading conditions, the peak current occurs at maximum load current plus maximum ripple. During an overload condition the peak current is limited to 4.2 A nominal (5.1 A maximum). The selected inductor (see Table 1) has a conservative 6.2 Amp saturation current rating. For this manufacturer, the saturation rating is defined as the current necessary for the inductance to reduce by 30%, at 20°C.

```
% minimum load current is 100mA
% 9V for headroom on Vin
L = ((3.3)*(9-3.3))/(0.5*500e3*9)
```

```
L = 8.3600e-06
```

```
%8.36uH inductor, closest is 4.2uH i think
```

```
% milli -micro- nano -pico
```

```
% https://dk.rs-online.com/web/p/smd-spoller/2317197
```

```
% the inductor above has Irms=5A, Isat=7.5A, metal alloy construction
```

```
% DCR = 64mOhm P = I^2rms * DCR
```

```
P=3^2*64e-3 % (same but add good copper around inductor for heat dissipation)
```

```
P = 0.5760
```

```
% SRF = 12MHz well above 500kHz (self resonant frequency)
```

```
% (rule of thumb SRF atleast 10x higher than Fs (it is)
```

```
% Fs *10 is 5Mhz
```

```
% Shielded metal-alloy core: good for low noise at 500 kHz.
```

```
% ramp capacitor With the inductor value selected,
```

```
% the value of C3 (CRAMP) necessary for the emulation ramp circuit is:
```

```
% use nominal inductor value (the theoretical closest real value)
```

```
% (suggested by TX)
```

```
Cramp = 8.2e-6*10^-5
```

```
Cramp = 8.2000e-11
```

% so 82 pF ceramic capacitor

% E12 series is 82 Pico Farad (choice) (in adafruit 441 book)

But in capacitors, the *common* series used depends on the tolerance and dielectric:

Type	Typical tolerance	Common series
Ceramic (NP0/C0G, X7R, X5R)	±5%, ±10%, ±20%	E6 or E12
Electrolytic / Tantalum	±20%, +80/-20%	E6 (coarse values)
Film caps	±5% or ±10%	E12 or E24

% Calculation of output capacitors

8.1.5 C9, C10

The output capacitors, C9 and C10, smooth the inductor ripple current and provide a source of charge for transient loading conditions. For this design a 22 µF ceramic capacitor and a 150 µF SP organic capacitor were selected. The ceramic capacitor provides ultra low ESR to reduce the output ripple voltage and noise spikes, while the SP capacitor provides a large bulk capacitance in a small volume for transient loading conditions. An approximation for the output ripple voltage is:

$$\Delta V_{OUT} = \Delta I_L \times \left(ESR + \frac{1}{8 \times F_S \times C_{OUT}} \right) \quad (9)$$

For a buck:

$$\Delta I_L = \frac{V_{OUT}(1 - \frac{V_{OUT}}{V_{IN}})}{L f_S}$$

$$\Delta I_{L_max} = (3.3 * (1 - (3.3/8.4))) / (8.2e-6 * 500e3)$$

$$\Delta I_{L_max} = 0.4887$$

$$\Delta I_{L_min} = (3.3 * (1 - (3.3/6))) / (8.2e-6 * 500e3)$$

$$\Delta I_{L_min} = 0.3622$$

% ESR = *equivilant series resistance*

If you use one 150 µF polymer (ESR ≈ 20 mΩ) in parallel with a 22 µF ceramic (ESR ≈ 3 mΩ):

- Total $C_{OUT} = 150 + 22 = 172 \mu F$
(for ceramics, you can optionally derate 20–40% for DC bias).
- Parallel ESR:

$$ESR_{eq} = \left(\frac{1}{ESR_{poly}} + \frac{1}{ESR_{cer}} \right)^{-1} \approx (1/0.02 + 1/0.003)^{-1} \approx 2.6 m\Omega$$

$$ESReq = (1/0.02 + 1/0.003)^{-1}$$

$$ESReq = 0.0026$$

```
% polymer + ceramic gives you very low ripple
```

```
% use 1 polymer as bulk cap with low esr and use a ceramic for
```

```
% polymer bulk cap https://de.farnell.com/en-DE/panasonic/eefsx0j151er/cap-150uf-6-3vdc-alu-elec-poly/dp/1751929
```

```
% ceramic cap
```

```
%DeltaVout =
```

```
% calculating R5, R6
```

R5 and R6 set the output voltage level, the ratio of these resistors is calculated from:

$$R5/R6 = (V_{OUT} / 1.225 \text{ V}) - 1 \quad (11)$$



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Application Information (continued)

For a 5 V output, the R5 and R6 ratio calculates to 3.082. The resistors should be chosen from standard value resistors, a good starting point is selection in the range of 1 k Ω - 10 k Ω . Values of 5.11 k Ω for R5, and 1.65 k Ω for R6 were selected.

```
rfive_div_rsix = (3.3/1.225)-1
```

```
rfive_div_rsix = 1.6939
```

```
% for 3.3V = 1.6939
```

```
rfive = ((3.3/1.225)-1)*3.3e3
```

```
rfive = 5.5898e+03
```

```
% r5 = 5.1k, r6=3.3k (in adafruit 441)
```

```
% calculating R1 R6 C12 (see initial schematic in datasheet)
```

8.1.12 R1, R2, C12

A voltage divider can be connected to the SD pin to set a minimum operating voltage $V_{IN(min)}$ for the regulator. If this feature is required, the easiest approach to select the divider resistor values is to select a value for R1 (between 10 kΩ and 100 kΩ recommended) then calculate R2 from:

$$R2 = 1.225 \times \left(\frac{R1}{V_{IN(min)} + (5 \times 10^{-6} \times R1) - 1.225} \right) \quad (12)$$

Capacitor C12 provides filtering for the divider. The voltage at the SD pin should never exceed 8 V, when using an external set-point divider it may be necessary to clamp the SD pin at high input voltage conditions. The reference design utilizes the full range of the LM25576-Q1 (6 V to 42 V); therefore these components can be omitted. With the SD pin open circuit the LM25576-Q1 responds once the V_{CC} UVLO threshold is satisfied.

```
% calculating R4 R5 and R6
```

```
Fpmod = 1/(2*pi*50e3*0.01e-6)
```

```
Fpmod = 318.3099
```

The dominant low frequency pole of the modulator is determined by the load resistance (R_{LOAD}) and capacitance (C_{OUT}). The corner frequency of this pole is:

$$f_{p(MOD)} = 1 / (2\pi R_{LOAD} C_{OUT})$$

For $R_{LOAD} = 5\Omega$ and $C_{OUT} = 177\mu F$ then $f_{p(MOD)} = 180\text{Hz}$

DC Gain_(MOD) = $2 \times 5 = 10 = 20\text{dB}$

For the design example of [Functional Block Diagram](#) the following modulator gain vs. frequency character was measured as shown in [Figure 15](#).